BSS310 SEMESTER PROJECT OVERVIEW AND TEMPLATED WRITING GUIDE

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Overview

The semester project is a summative component of the module with two outcomes viz: group work and individual work. It focuses on addressing ECSA GA11 (Engineering Management) from an applied-theoretical point of view, where students need to demonstrate competence in three (3) outcomes, viz.: semester project [(i)**Group work**, (ii) **Individual work**] and (iii) the written **Examination** to pass the module. In a bid to expose students to entrepreneurship, project, and financial management, this module, as stipulated by ECSA GA11, explored these areas from a business point of view. The students are expected to set up their own business-based project as an interdisciplinary team from different departments, in groups of three students.

Based on this, the focus themes for the module span: business systems design and management (BSDM), project management (PM), and economic decision-making (EDM). Students are mostly advised to identify one or two modules in their departments and see which of these modules can be fine-tuned and developed into a useful practical knowledge domain, and evolve into a business. Alternatively, students are allowed to explore projects not directly linked to their course of study; however, in either scenario, they should be able to apply the theoretical concepts of strategic market analysis to aid them in their transition from a congested red ocean to a unique blue ocean scenario.

This short piece provides a template for the semester project by exploring the different sections of the rubric. The document is divided into two parts covering several sections. These include the **Group Work,** addressing **Part A,** and the **Individual Work,** covering **Part B**. Both parts have a couple of inherent sections and sub-sections.

PART A TEMPLATE (GROUP WORK)

An Overview

Part A starts the project by clearly stating the problem. As per the guidelines set in the study guide, a gap in the business system needs to be identified, and a need conceived. The need could be **a product or a service** that is conceptually designed and developed to address the identified gap. It is the responsibility of the group to decide whether their semester project will be focused on **a product or a service industry**.

Based on past observations in the module, it is strongly recommended that groups avoid working on a product/service that is too complex to be explored and finalised within the stipulated duration. An example of a product considered to be “too complex” can be the building of a new space shuttle, while an example of a service system considered to be too complex can be the design and development of a new software solution. **It is advisable to choose a product or a service that can be split into a series of activities and tasks, and is realisable at the end, to avoid making the project unnecessarily difficult to execute.** An example of a realisable product can be a “**silent kettle boiler with a smart heating-voltage regulatory ability**”, hence you can choose when you want your water boiled, etc. An example of a realisable service can be “**a car wash business with an integrated service to collect clients' cars from their homes or offices (optional), provided the distance is within 10kms; and auto-notification when washing is completed plus delivery service (optional)**”. No need to stay in the queue in-person.

Nevertheless, in as much as we desire to avoid any form of restrictions in the choice of business systems, if you find yourself in an assumed “complex project space”, clearly define your mission; what you wish to pursue now such as segment one of the hyper complex project while stating what the likely segments two, three etc would be in your report. Ensure that “segment one” is a project segment with finality and sellable as a “valuable offering” to clients and not an incomplete work in-progress.

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Section 1: Problem identification

Write a clear problem statement

- Firstly, it is pertinent that we know the sector you are concerned with as you start with the problem motivation e.g. "**The banking sector in South Africa has seen a drastic decline in the use of chatbot by its clients due to unreliability of the assumed smart platform**". From this, it is clear you are concerned with the banking sector. A problem statement is basically a statement of negations, mishaps, challenges, etc., in a given knowledge domain. This can range from the service to the product industries; however, you are expected to pick only one sector. A problem statement should be very specific and clearly defined within the chosen area. The problem statement is all about gap identification and analysis. A precise question such as: what is the gap in this sector, should be asked for a start. Use if available in the literature, past data from reports, articles or research papers to motivate the gap).

To complement the problem identification, the analysis of the problem can be explored i.e. the extent of the problem, the spread of the problem, the complexity level of the problem, who is affected by the problem and to what extent, whether it is local or international or both etc can be further explored, why it is important to address the identified gap in the chosen sector, what will happen in terms of negative effects if this gap is not addressed etc. Possible infographics and/or statistical data to further buttress the existence of the identified problem can be presented within the writing limits.

Section 2: Objectives and Operational Requirements

In this part of the report, the group must successfully demonstrate clear use of techniques from BSDM and PM. It is expected that the group carry out the following:

- Identify the business system's requirements in terms of:

i. objectives: (State the goals of your business system; between three (3) to five (5) goals can be considered appropriate and viable). Tree diagrams are allowed here as deemed necessary.

ii. system's operational drivers: (Identify those factors that are considered to be necessary for the business to kick start. Briefly motivate on these factors).

- Show what capabilities (functions) the system needs to have in place to achieve the stated objectives: To achieve the stated goals, some enablements or capabilities must be in place. These enablements must be linked or connected to at least one of the stated objectives. However, you can have two or more functions linked to one objective and/or a function linked to two or more objectives. Tree diagrams are allowed here as deemed necessary. Be yourself and be sure of what you are doing.
- Identify physical embodiments for the functions: All functions must have a hardware or software element that encapsulates them. No functional propagation is possible without a hardware or software embodiment. Link the stated functions to specific embodiments.
- Validate the business system's needs: (You are expected as a group to explore, link, and discuss the trio concepts: Measure of Effectiveness (MOE), Measure of Performance (MOP),

Technical Performance Measure (TPM). Motivate on how the different segments or levels (systemic level, sub systemic level and component level) of the business can be traced and tracked for non-performance, monitoring for maintenance and improvement, etc. Firstly, the systemic level is characterised by the MOE concept, followed by the MOP and TPM concepts as presented below.

- Measure of Effectiveness (MOE): This validation procedure focuses on the system as a whole. Motivate on how you can tell that the problem or growth is coming from a particular, generic functional attribute or attributes of the business, e.g., late staff arrival, outage of power, delivery of poor services, etc. These functional observations cannot be initially traced or linked to any of the subsystems, i.e., any of the portfolios or departments, without carrying out a thorough check, hence making it a system-level observation and thinking. At the systemic level, you don't carry out any checks, you only observe based on the set out functional and physical attributes.

- Measure of Performance (MOP): This validation procedure focuses on the sub-systemic level, i.e., the different portfolios or departments, after observing the following at the systemic level, i.e., late staff arrival, outage of power, delivery of poor services, etc. However, you need to motivate on how you can tell that the problem picked up or the growth recorded is coming from the production, marketing, or logistics department, or all three departments, etc. This must be done with specific measurable functional and physical attributes.

- Technical Performance Measure (TPM): This validation procedure focuses on the component level, which is the sub-sub-systemic level. Herein, it is certain which portfolio is the focus. However, you need to be sure of the specific staff, specific machine, specific policy, etc., within this portfolio that is resulting in the positivity or negativity.

Hence, in summary, validation seeks to state that your business has in place all of these tracking capabilities, so when things go wrong or go in the positive direction, it doesn't appear as a mystery. Ensure coherence across the entire section. It is strongly recommended that the group motivate each point in a strong manner that indicates a reasonable depth of knowledge. As a rule of thumb, back every idea/content you present as a group with knowledge acquired from the module's lectures.

Section 3: Market Analysis

In this section, the group is expected to successfully perform a market analysis by applying the techniques from EDM as presented in the lectures. The following content is expected in this section of the report:

- Define the target market: A clear and detailed target market should be listed and discussed based on a prior segmentation and analysis premised on SWOT and PESTLE analysis. For instance, how did you conclude that this cluster of consumers is your target market? Any previous segmentation and targeting carried out, etc.? Was there any previous market campaign carried out, etc.? Did you consider the possible government political will, economic climate of the nation, social and cultural demands, technology, legal implications, and environmental impact of the business? Did you explore your attributes, such as strengths, weaknesses, etc.?

- Complete a marketing map to identify the gap: A marketing map for positioning analysis should be presented and explored; this would likely uncover the degree of competitiveness in that space and how the potential or actual clients currently see the organisation. It's a way to uncover strategic gaps in the sales of a particular service or goods. You want to visualise how your business currently sits with at least four competitors in the business domain.
- Use of Porter's five forces as a framework to discuss competitive forces: (Each of the 5 forces considered by Porter must be clearly explained in depth, bearing in mind how these are linked to the current business).
- Stating a unique value proposition: (The unique value proposition is a clear statement that speaks to the specifics or uniqueness you wish to bring to your business; this should be capable of transitioning your business from the red to the blue ocean or sustaining your business right in the blue ocean, as the case may be.

In addition to the above-mentioned, the group can utilise credible reference sources as proof to support their in-depth market analysis.

Section 4: Value Chain Design

- Describe a high-level design for each of the primary and support activities in your value chain (1 paragraph each): You are expected to present how your value chain was designed. And how you reached the "unique value proposition" as presented in section 3.

In using Porter's value chain, you are required to identify up to three (3) or more primary activities (covering the direct and indirect activities), up to three (3) or more support activities (covering the direct and indirect activities) of your business. Note that the direct and indirect activities are the corresponding sub-activities within the primary and support activities clusters. The linkage amongst these should be clearly stated and reviewed in a bid to track possible nonperforming sub-activities, including a need to prioritise the sub-activities based on dependence on each other.

Furthermore, a view of the SIPOC concept as explored in the lecture towards addressing this subsection in your business would be appreciated.

The "**value proposition**" presented in section 3 under market analysis should be strongly linked to the breakdown in this section, i.e., the sub-activities presented herein. The competitive advantage of the value proposition should be provable and coherent with this section, presenting the value chain design.

After section 4, you are expected to branch out to the Part B rubric of the Semester Project for the individual task.

The following sections 5 & 6 of Part A will now focus on two sections, viz.: "overall project plan" and "overall project cash flow analysis". Initial inputs to these sections are based on the works from Part B of the rubric covering the individual portfolios, i.e., (Production, Logistics & Marketing).

Section 5: Overall project plan (based on PART B)

In this section of the report, the following four bullet points are expected to be addressed by the group:

- Presentation of a high-level network diagram that shows the critical path (a minimum of 15 tasks & maximum of 20 tasks). (A minimum of 15 activities from all the portfolios implies 5 activities per portfolio, etc. To pick five (5) activities from the network diagram of your portfolio in Part B, you may decide to collapse 2, 3, 4, or more activities (from the main network diagram of your Part B) into one activity and assign a new generic name that covers all the activities that were collapsed. The new activities should be linkable to the main network diagram via the use of a descriptive legend as a pointer. Hence, with this approach, you can create and extract five or more new activities **across the different phases of the main network diagram in your respective Part Bs**.

You are meant to export the newly created activities, of course, to Part A here!!! This should be done for all three (3) portfolios, or two (2) portfolios in the case of a group of two (2).

Form a new network diagram here (in this section) based on the exported activities from the different portfolios.

- Systems thinking & business structure integratedness: You will be introduced to a free software, Vensim, for systems thinking. You want to see all the activities in the new network diagram as an integrated entity. You want to understand and understudy how these activities influence each other and whether the effect of this influence is positive or negative. This can aid in a rethink of the whole connectivity if the "impact of one activity on the other is negative and detrimental to the business. Also, the indirect impact of activities can be studied. All of this will be based on the consolidation of Part B activities.
- Present a consolidated risk matrix with a brief description of a mitigation strategy for each: The group should establish a detailed risk matrix, including mitigation strategies, and discuss how this will play out in the business. The chosen risk mitigation strategies must be coherent with the identified risks, including their respective positions on the matrix.
- Business intelligence: You are expected to present a simple machine learning model to look ahead in terms of the predictability of the business horizon. Trying to see if the introduction of new variables can be a game-changer for the near future, etc. Hence, a simple Naïve Bayes' theory would be adopted for this module and report writing.
- Business Model Canvas (BMC): You are required to create a business model canvas (BMC) based on inputs from the individual portfolios in section 3 of Part B. In this subsection, you are expected to strategically synchronise all Part B inputs here. A BMC is a concise executive summary of the business strategic plans, both functionally and operationally, with nine (9) components.

In a nutshell, section 5 is all about project management and the use of project-based activities to smartly anticipate the behaviour of a business yet to be operated, which is termed "**business intelligence**". Also, contained in this section is the Business Model Canvas that gives a condensed view of the business's functional and operational structure.

Section 6: Overall Project Cash Flow Analysis plan (based on PART B)

In this section of the report, the techniques that will be required from the group will come from the knowledge of EDM. The following is expected from the group:

- Present consolidated cash flow for the first three years of operation: (The expectation here is to operate the business on paper, for a duration of three years, by using a spreadsheet or balance sheet approach where cost and revenue meet for the evaluation of accumulated profit). This can give an idea of the amount of capital required to kick-start the business.
- Define and quantify worst case, best case, and most likely cash flow projections: (You are expected to play around with varied interest rates (low & high) about the current interest rate. Also, play around with the higher cost of goods and services at the peak level versus the lower cost of goods and services. This will aid in the generation of new consolidated cash flow tables and new profit margins.
- Determine the MARR for this project with substantiation: (The cost of capital is to be determined by the team. Capital sourcing for businesses can be split across the two possibilities i.e. debt or equity sources. Work towards a cheaper cost of capital.
- Conduct a Net Present Value (NPV) analysis and make recommendations regarding the financial feasibility of the project: (Carry out a time value of money analysis by conducting a Net Present Value (NPV) analysis and make recommendations regarding the financial feasibility of the business over the given time horizon).

There should be a strong and clear correlation between the overall plan and the content from part B of the semester project. The correct interpretations must be made based on data/numbers/figures generated in that section of the project. Groups are encouraged to avoid any visible errors. The MARR calculated for this project should be accompanied by good reasoning around the market and business as a whole, by the group.

Conclusion (Optional): Let us know if the business is worth investing in, if it is worth the effort, and if it is viable. Some businesses may not be viable for the stipulated 3years monitoring period for diverse reasons. Let us know if yours falls in this category. A short, conclusive remark is required based on this. No wrong or right conclusion, provided the technical content and analysis are aligned with the concluding decision.

NB: The following are excluded from the 20-page max. count.

References

Appendices

Group Meetings

Furthermore, in Part A, the writing "professionalism" should be adhered to as highlighted below:

Professionalism: Part A: max. page limit: 20 pages (excl. Title page, project work declaration, executive summary, table of contents, list of figures, list of tables, labelled tables at the top of the table, labelled figures underneath, references for in-text citations, list of acronyms, reference list, meeting minutes' appendix. minimum font size 11pt, minimum margins 1cm; line spacing 1,15, font types can be Arial, Times New Roman, or Calibri.

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